

Tree terms have candidate exponents:
every term of the Taylor tree is $\sim C n^{-\mu}(\log n)^j$ with $\mu \in \Lambda(h, k)$,
 $j \leq d - 1$

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Abstract

The proof of `thm:TaylorTree` expands the standard integral into tree terms $\int_{(0,b]^d} u^{h+s} (\sqrt{n} u^k)^p e^{-\beta n u^{2k} + \beta \sqrt{n} u^k \xi(0)} du$, indexed by a shift $s = \mathbf{m} + \mathbf{n} \in \mathbb{N}^d$ and an insertion order p . Each is $n^{p/2}$ times a box integral with exponents $h + s + pk$, so unit 59 gives $\sim C n^{-\mu}(\log n)^j$ with $\mu = \min_i (h_i + s_i + 1)/(2k_i) \in \Lambda(h, k) = \bigcup_i ((h_i + 1)/(2k_i) + \mathbb{N}/(2k_i))$ and $j \leq d - 1$: the exponent set and the log-degree bound of the theorem, term by term. The insertion order does not move the leading exponent. Zero `sorry/axiom`.

1 Setting

The insertion $(\sqrt{n} u^k)^p$ is $n^{p/2} \prod_i u_i^{pk_i}$, which raises every candidate exponent by $p/2$ and is exactly compensated by the prefactor $n^{p/2}$; the shift s moves the exponents along the arithmetic progressions of $\Lambda(h, k)$.

2 The things formalised

```
def candidateSet {d : } (k h : Fin d → ) : Set :=
  { | (i : Fin d) (s : ), = ((h i : ) + s + 1) / (2 * k i)}
noncomputable def treeTerm ( a b c : ) {d : } (k h s : Fin d → ) (p : ) : :=
  u : Fin d → , ( i, u i ^ (h i + s i)) * (c * i, u i ^ k i) ^ p
  * quadKernel a (c * i, u i ^ k i) (boxMeasure b d)
theorem treeTerm_eq ... :
  treeTerm a b c k h s p
  = c ^ p * boxIntegralFin a b c k (fun i => h i + s i + k i * p)
theorem treeTerm_isEquivalent ( a b : ) (h : 0 < ) (hb : 0 < b) (D : )
  (k h s : Fin (D + 2) → ) (hk : i, 0 < k i) (p : ) :
  ( : ) (j : ) (C : ), candidateSet k h j D + 1 0 < C
  (fun n => treeTerm a b (Real.sqrt n) k h s p) ~[atTop]
  fun n => C * (n ^ (-) * Real.log n ^ j)
```

3 Proof strategy

The insertion identity is pointwise algebra (`pow_add`, `pow_mul`, `Finset.prod_mul_distrib`, `Finset.prod_pow`). For the asymptotic, unit 59 applied to the shifted exponents provides p' , m , C ; an index i_0 attaining the minimum exists because the attaining set has cardinality $m + 1 > 0$, and

$p' = (h_{i_0} + s_{i_0} + 1)/k_{i_0} + p$ by definition of the shifted exponent. Multiplying by $(\sqrt{n})^p = n^{p/2}$ and combining powers gives $n^{-\mu}$ with $\mu = (h_{i_0} + s_{i_0} + 1)/(2k_{i_0})$; $j = m \leq D+1$ since $m+1 \leq \#\text{Fin}(D+2)$.

4 Roadblocks and resolutions

- `Finset.prod_pow` rewrites $\prod f^p$ to $(\prod f)^p$; stating the shifted exponent as $k_i \cdot p$ (not $p \cdot k_i$) makes `pow_mul` produce the matching shape.
- Precedence of unary minus: the goal had $-(x/2)$ while the rewrite was stated as $(-x)/2$.

5 What was Mathlib, what was new

Mathlib: `Finset.prod_pow`, `Finset.card_pos`, `Finset.card_filter_le`, `Real.rpow_sub`. New: the candidate set, tree terms, the insertion identity, and the term-wise exponent theorem.

6 Lessons

Once the box theory is hypothesis-free, the combinatorial statements of the paper about which exponents can occur become one-line consequences; this is the right level at which to connect to the tree expansion, before the analytic questions about summing over s and p .

7 Outcome

Term by term, the Taylor tree expansion has exponents in $\Lambda(h, k)$ and logarithmic degree at most $d - 1$, with each leading coefficient positive. Summing the tree (absolute convergence, the $O(e^{-\varepsilon n})$ remainder, and the coefficients $C_{\mu, m}$ as series) is the remaining analytic content of `thm:TaylorTree`.